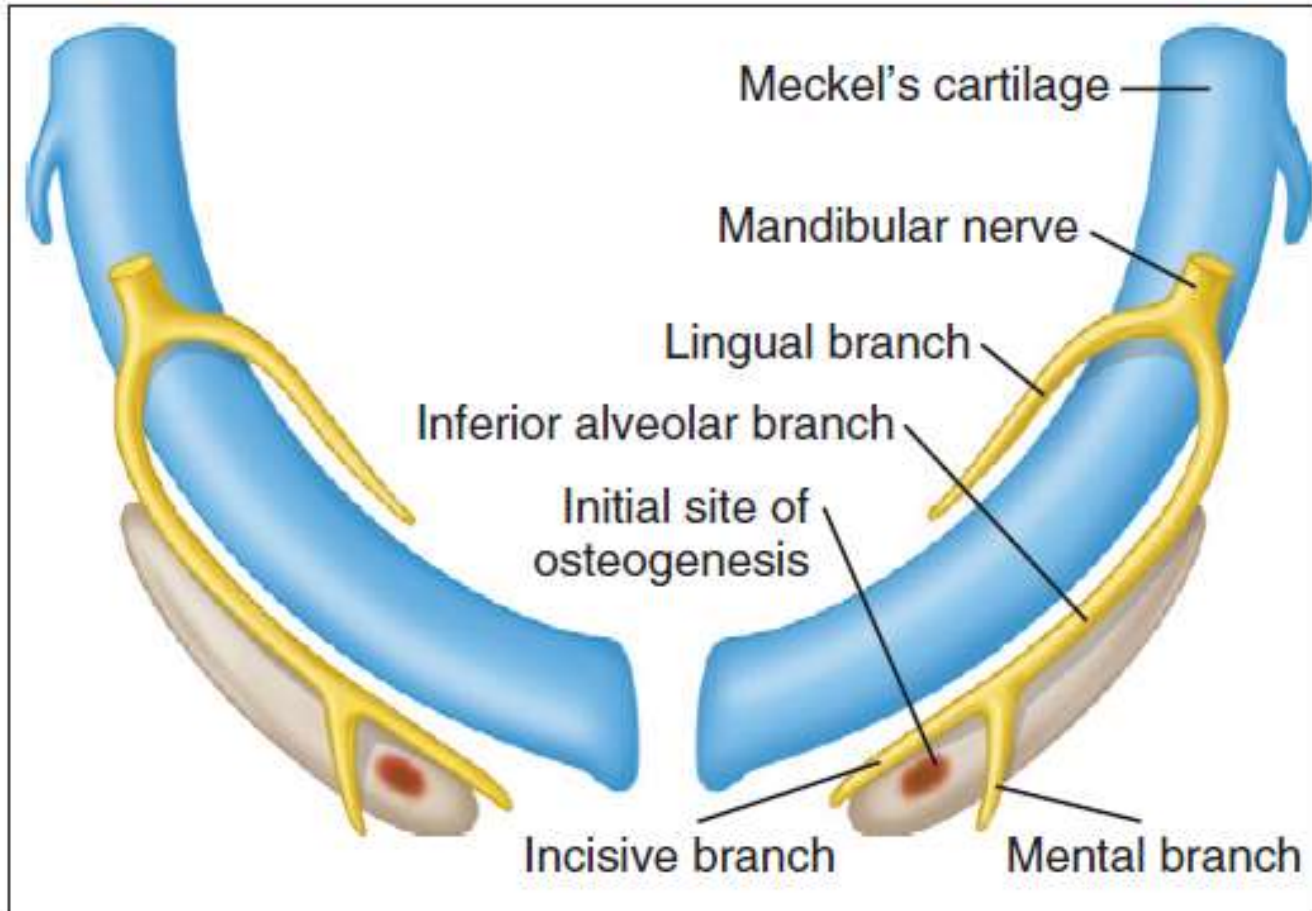


Development of Jaws

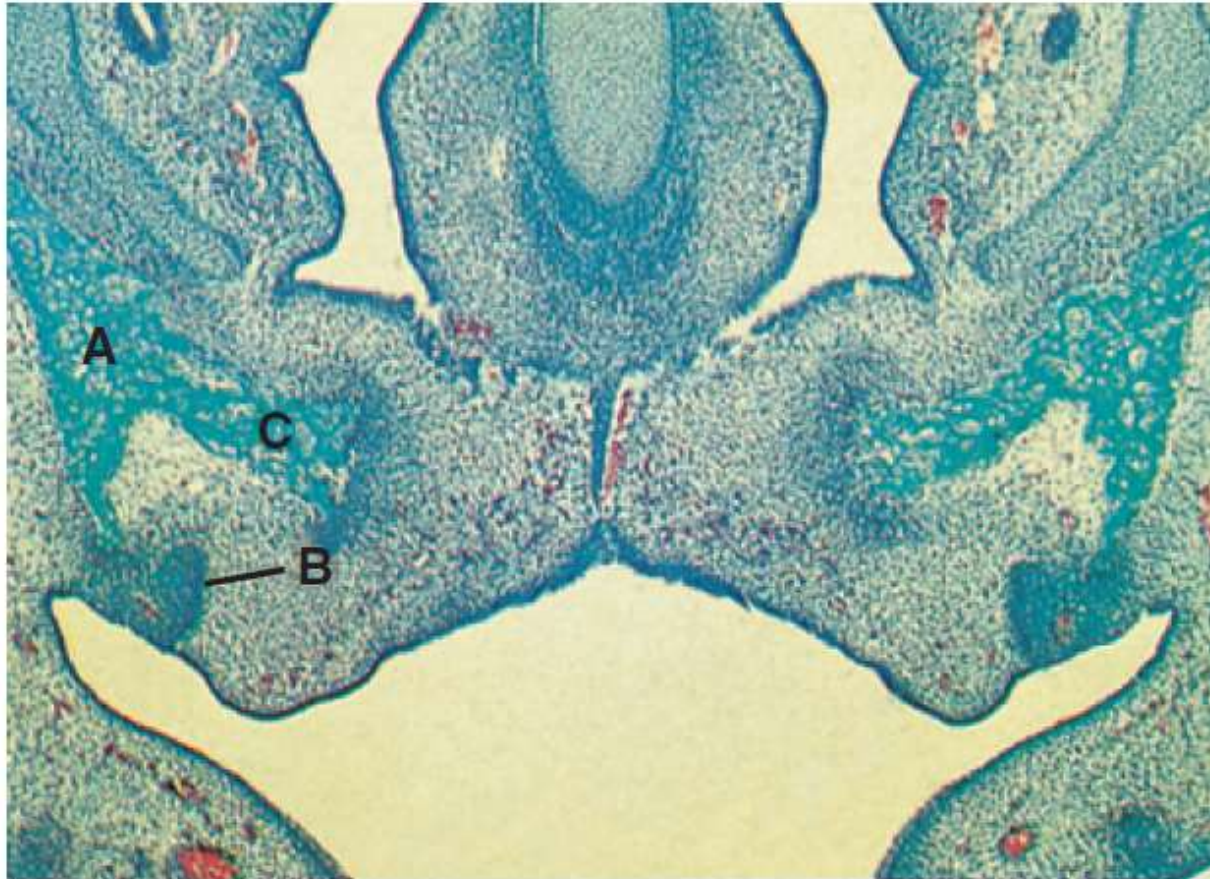
Dr. Ume Hani

Dept. of Oral Biology

TASK 1: IDENTIFY, STUDY AND PASTE THE DIAGRAMS



**DIAGRAMMATIC REPRESENTATION OF
EARLY DEVELOPING MANDIBLE:
Shown is site of initial osteogenesis
related to mandible formation**



CORONAL SECTION THROUGH DEVELOPING ORONASAL REGIONS SHOWING EARLY DEVELOPING MAXILLA:

A: the region of developing maxilla

B: The region of developing canine (site of ossification)

C: palatine process

TASK 2: ANSWER THE QUESTIONS

1- What is the fate of Meckel's Cartilage?

Although Meckel's cartilage contributes no significant tissue to the developing mandible, nodular remnants of cartilage may be seen in the region of the mandibular symphysis until birth and, in its most dorsal part, Meckel's cartilage ossifies to form ear ossicles (the malleus and incus). Behind the body of the mandible the perichondrium of Meckel's cartilage persists as the sphenomandibular and sphenomalleolar ligaments. The sphenomandibular ligament ossifies at its sites of attachment to form the lingula of the mandible and the spine of the sphenoid bone.

2- Name and explain the three secondary cartilages of mandible?

Between the 10th and 14th weeks in utero, three secondary cartilages develop within the growing mandible.

Condylar Cartilage: The largest and most important of these, appears beneath the fibrous articular layer of the future condyle. By proliferation and subsequent ossification the cartilage is thought by some to serve as an important centre of growth for the mandible functioning up to about the 16th year of life.

Coronoid Cartilage: A transitory secondary cartilage are seen associated with the coronoid process.

Symphyseal cartilage: A transitory cartilage found in the region of the mandibular symphysis.

3- Differentiate the intramembranous ossification from endochondreal

INTRAMEMBRANOUS OSSIFICATION

Mesenchymal tissue directly converts into bone.

Osteoblasts differentiate directly from mesenchymal cells.

Bone formation begins within mesenchymal condensations.

Forms most flat bones.

Examples: frontal, parietal, maxilla, most of mandible.

Ossification centers develop within mesenchyme.

Begins during fetal life and continues through growth and remodeling.

Primary mechanism for formation of craniofacial bones.

ENDOCHONDRAL OSSIFICATION

Bone develops by replacing a hyaline cartilage model.

Chondroblasts first form cartilage, which is later replaced by bone.

Bone formation begins within a cartilage template.

Forms most long bones and irregular bones.

Examples: femur, tibia, humerus, vertebrae, base of skull.

Primary and secondary ossification centers develop within cartilage.

Continues until epiphyseal plate closure at puberty/early adulthood.

Primary mechanism for formation of appendicular skeleton.

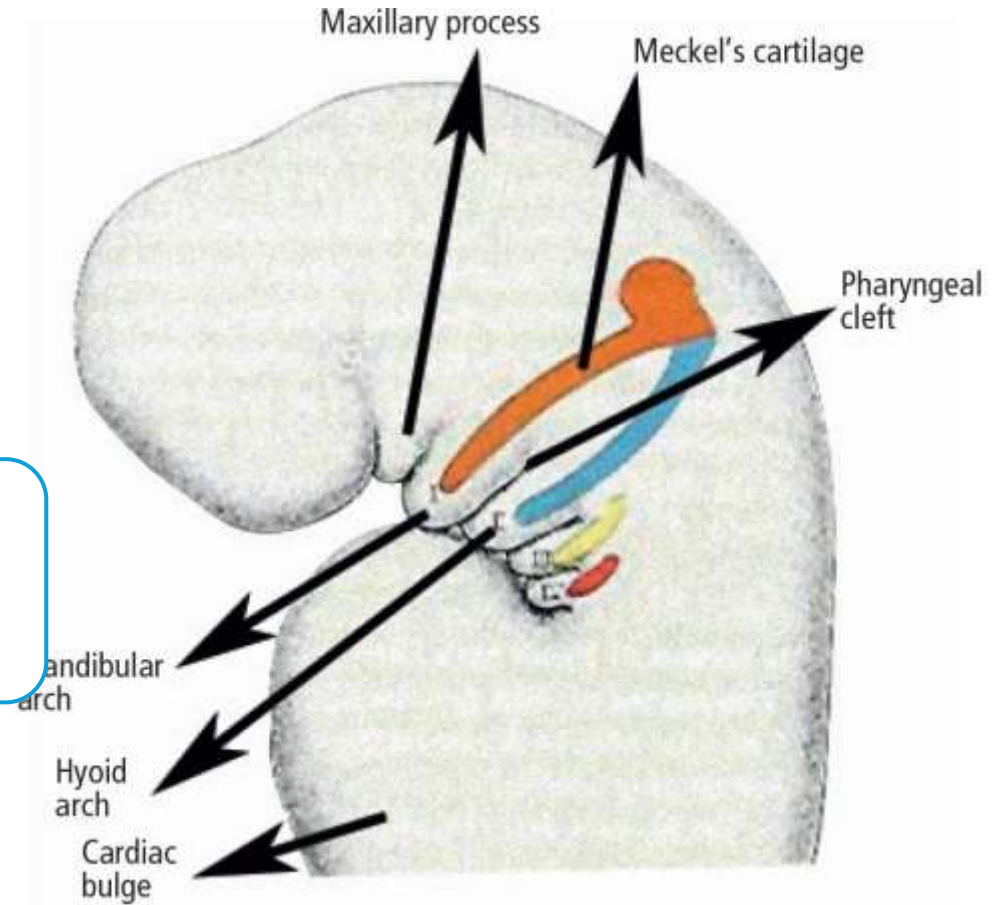
REFERENCE/STUDY MATERIAL

Development of Mandible

The mandible develops **intramembranously** from the first pharyngeal arch (mandibular arch) at **6th week**.

Meckel's cartilage extends from the cartilaginous otic capsule in the region of the developing ear to a midline symphysis.

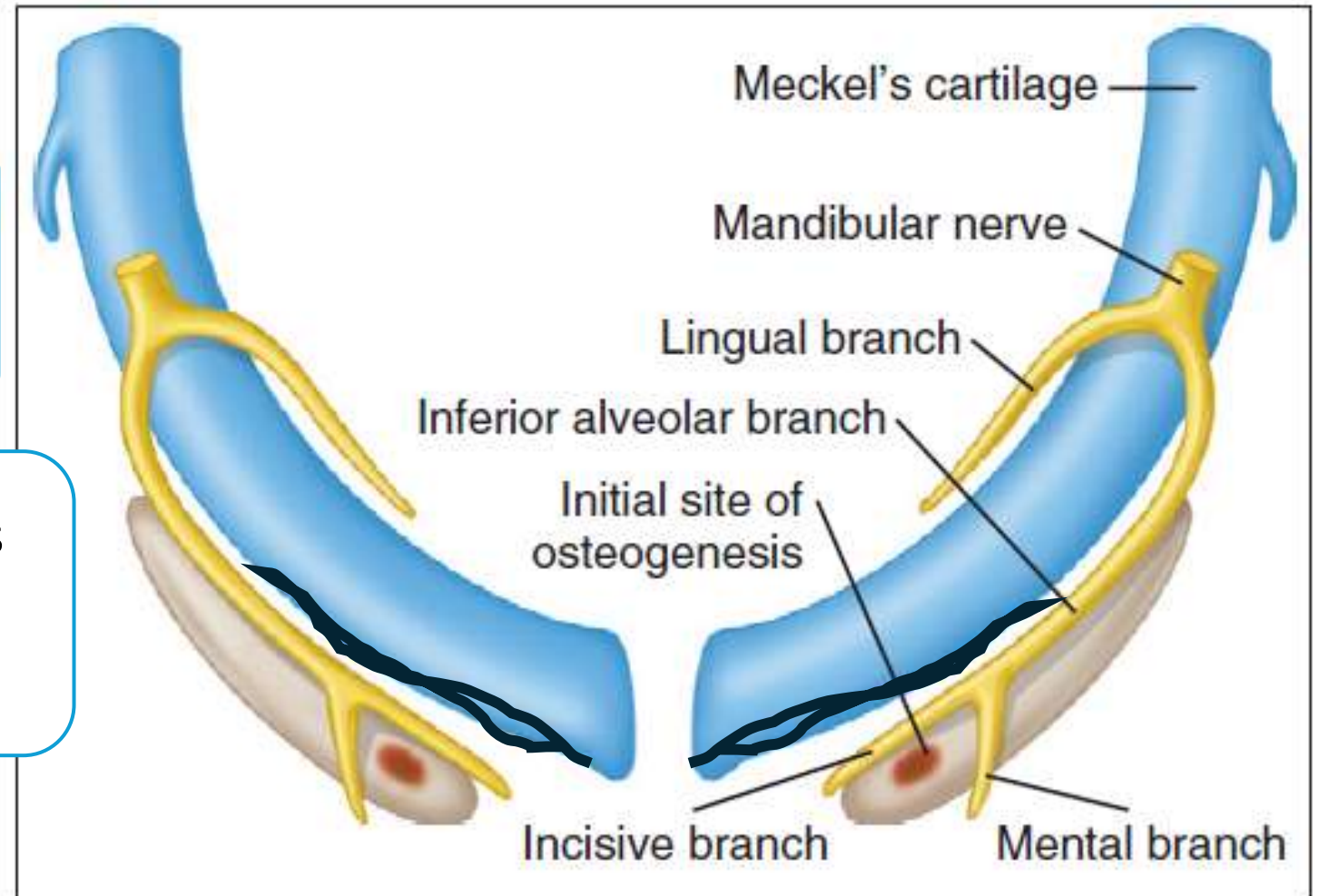
Meckel's cartilage **ONLY** provides a framework for the developing mandible.



Development of Mandible

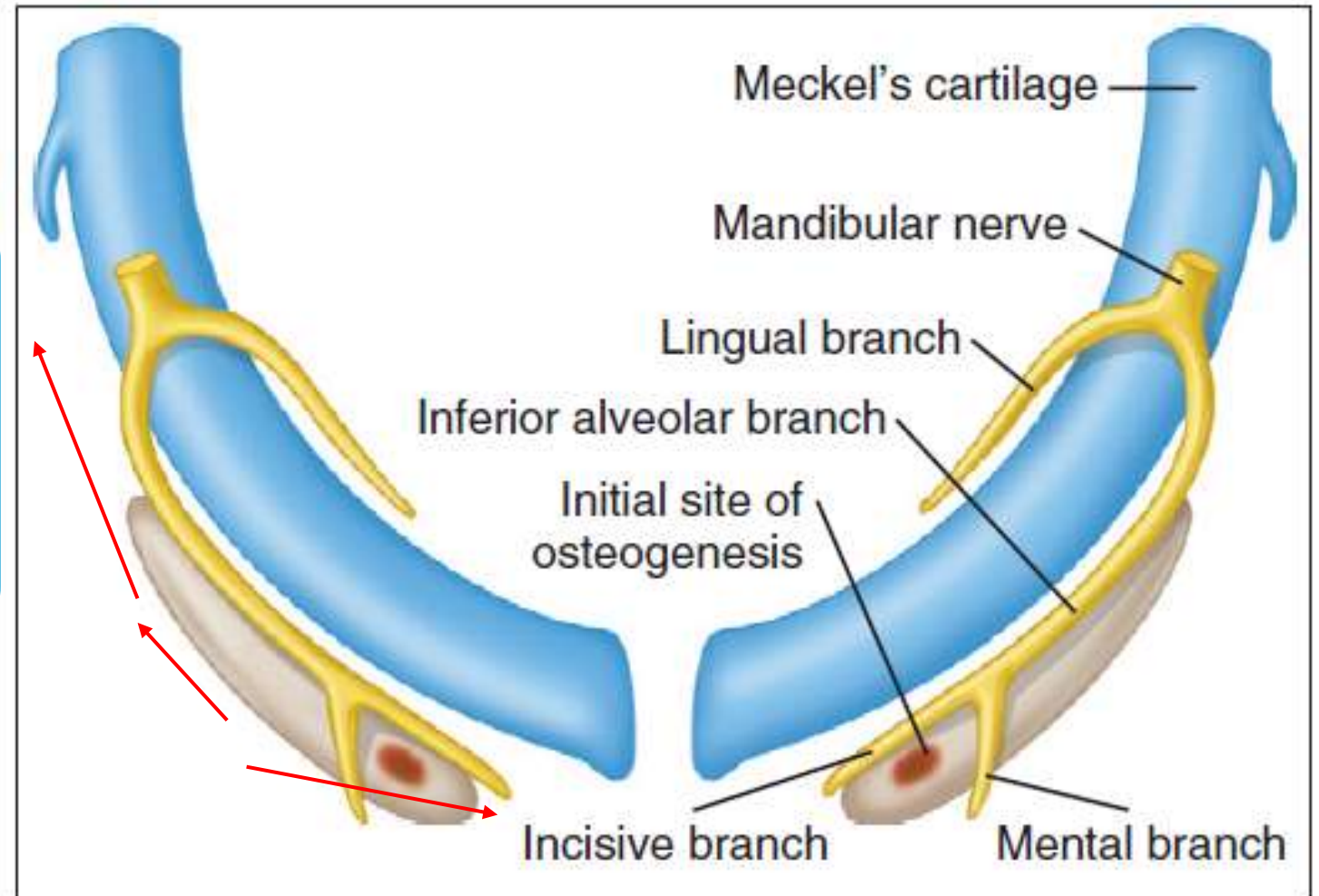
During the **7th week**, mesenchyme (anterolaterally) around Meckel's cartilage condenses to form fibrous tissue.

Centers of ossification appear in this fibrous tissue at a site close to the **future mental foramen**.



Development of Mandible

From this centre, bone formation spreads rapidly **backwards**, **forwards and upwards**, around the inferior alveolar, incisive and mental nerves

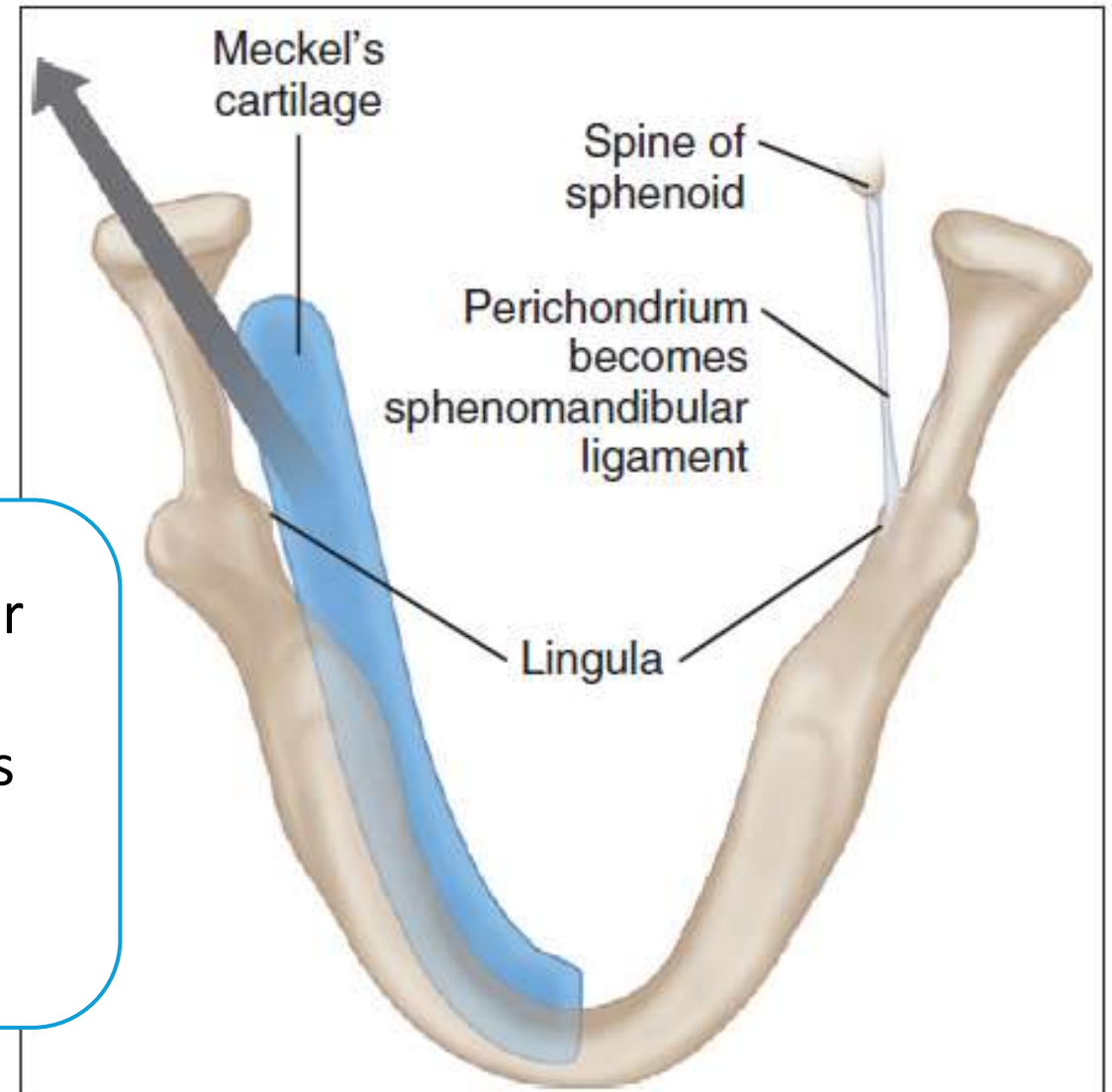


Development of Mandible

The ramus of the mandible develops by a rapid spread of ossification, posteriorly turning away from Meckel's cartilage.

Fate of Meckel's

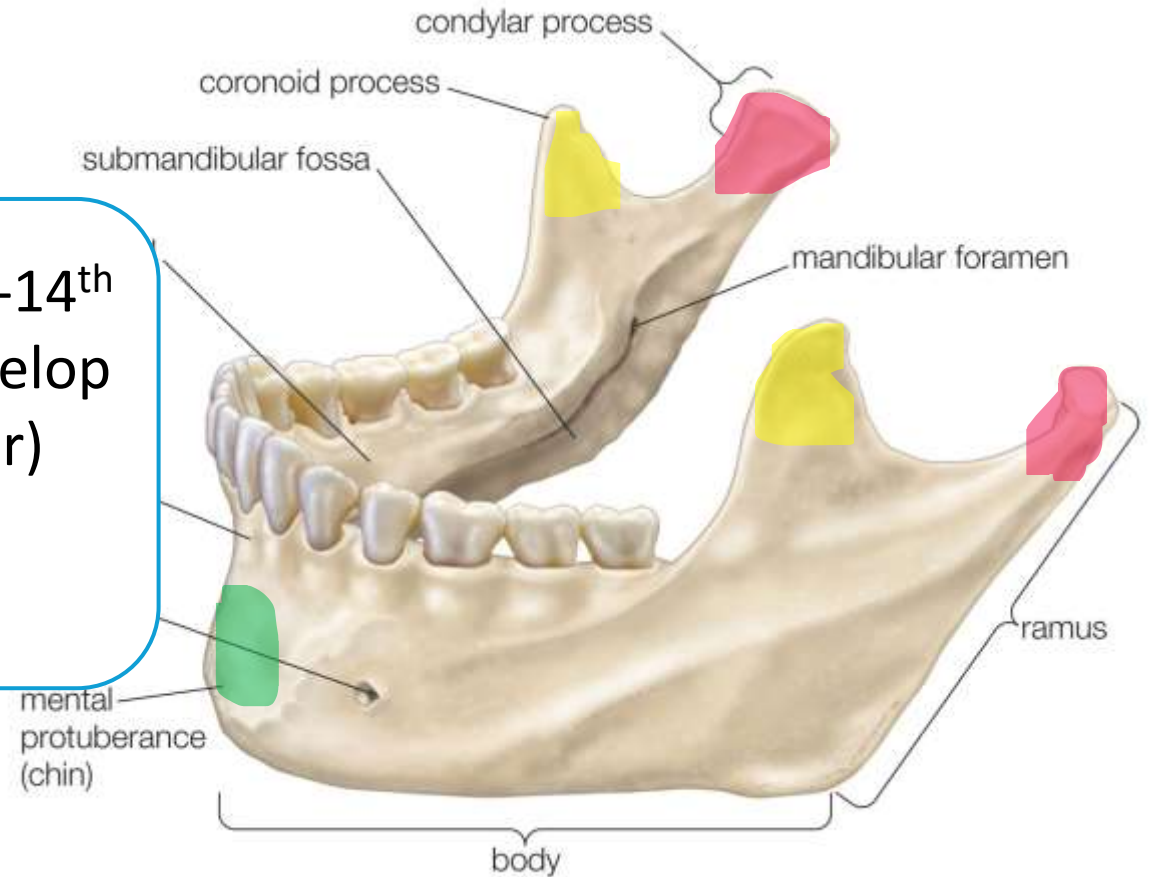
- remnants anterior region of the mandibular symphysis until birth
- The dorsal part ossifies to form ear ossicles
- Perichondrium of Meckel's persists as the sphenomandibular and sphenomalleolar ligaments



Development of Mandible

Endochondral ossification starts at 10-14th weeks when secondary cartilages develop

1. Condylar cartilage (growth center)
2. Coronoid cartilage
3. Symphyseal cartilage



Growth of Mandible

1. Growth of alveolar processes
2. Continuous remodeling
3. Condylar growth center (forward and downward)

Growth of Mandible

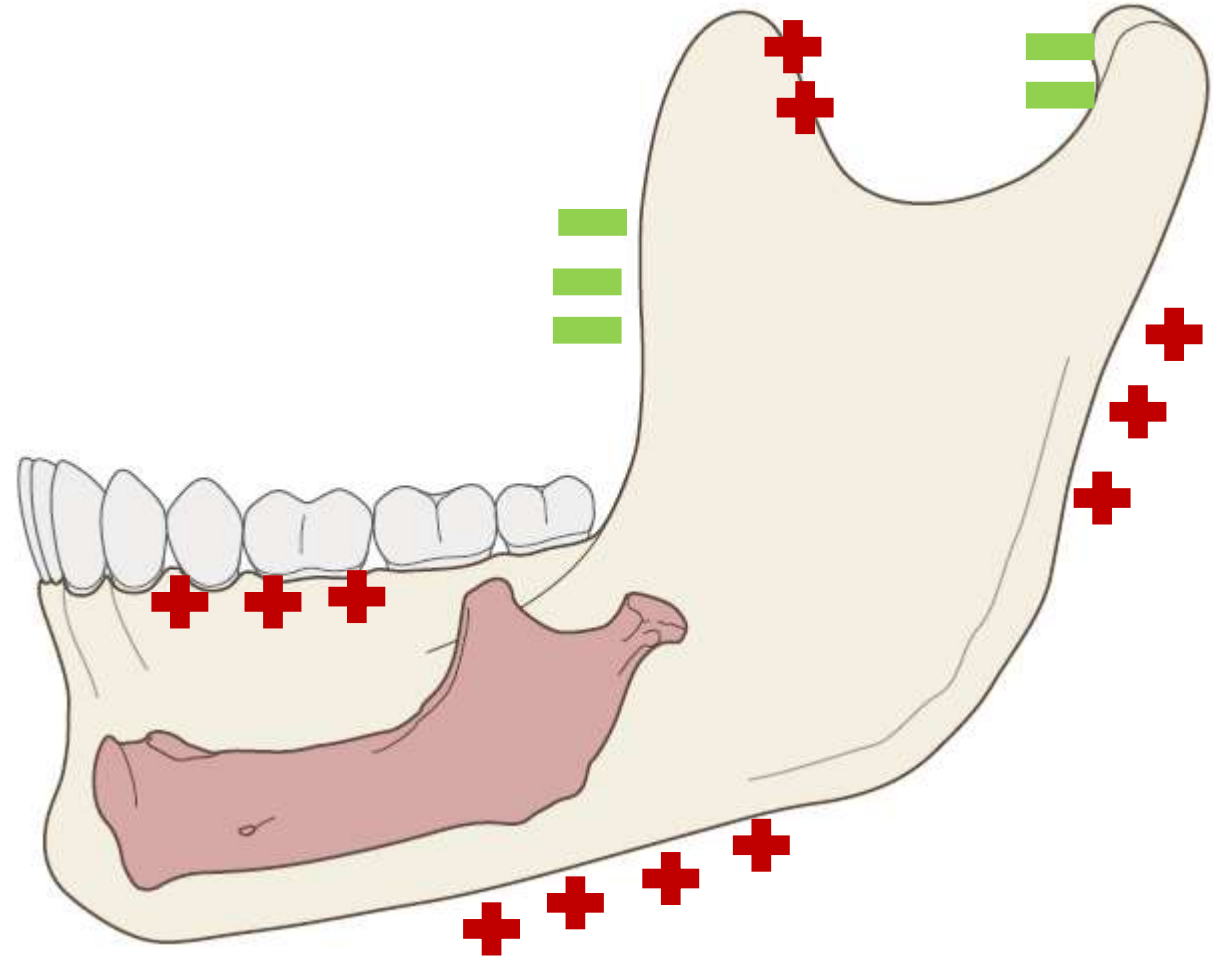
+ Deposition

— Resorption

Height: Alveolar processes

Length: Ramus, condyle and coronoid

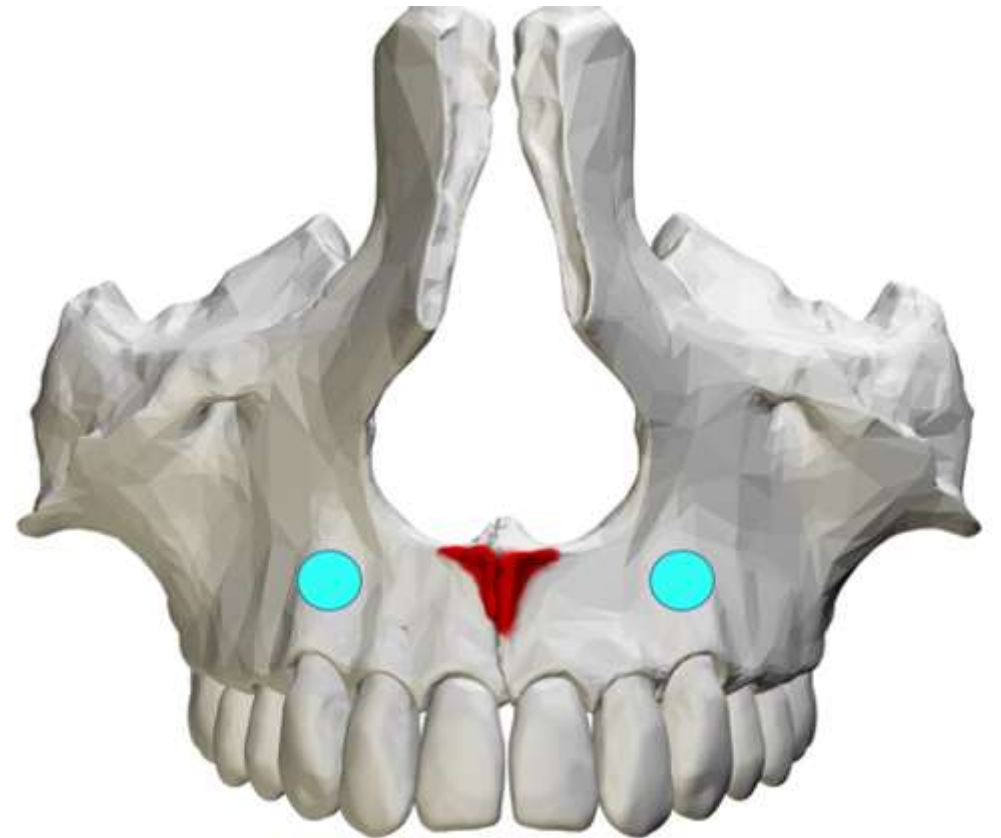
Width: Inner and outer surface of mandible



Development of Maxilla

The maxilla develops entirely by **intramembranous ossification** from the first pharyngeal arch.

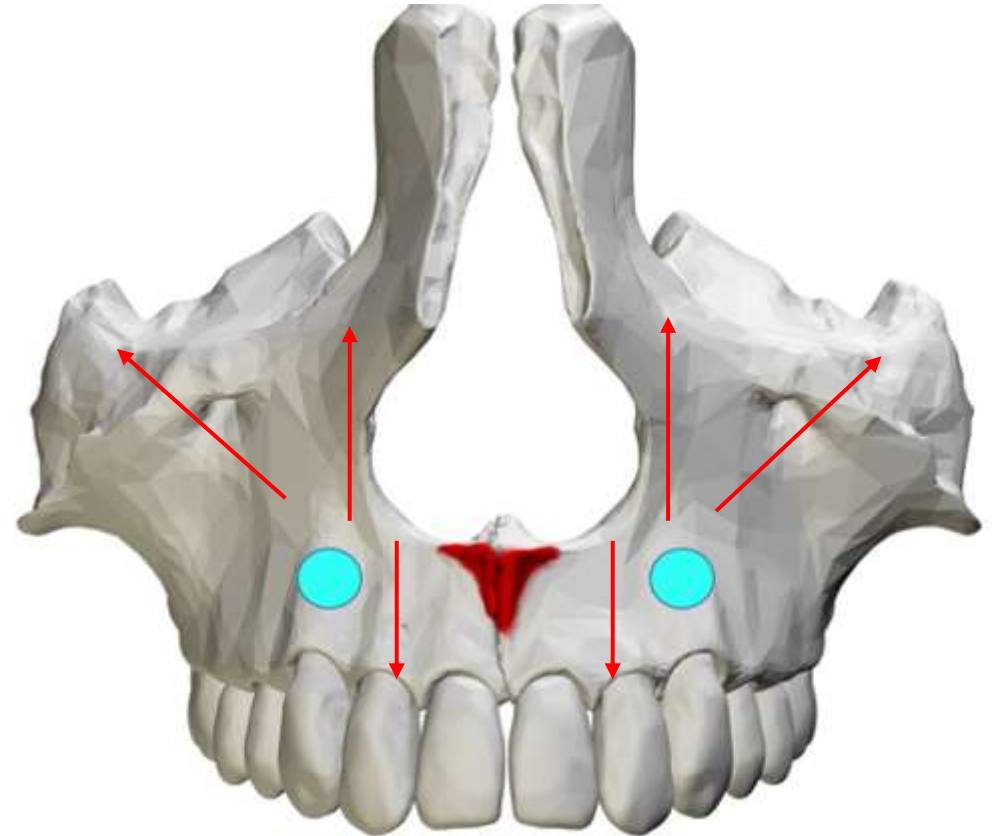
Develops in the mesenchyme of the maxillary process of the arch (developing deciduous canine) at **8th week**.



Development of Maxilla

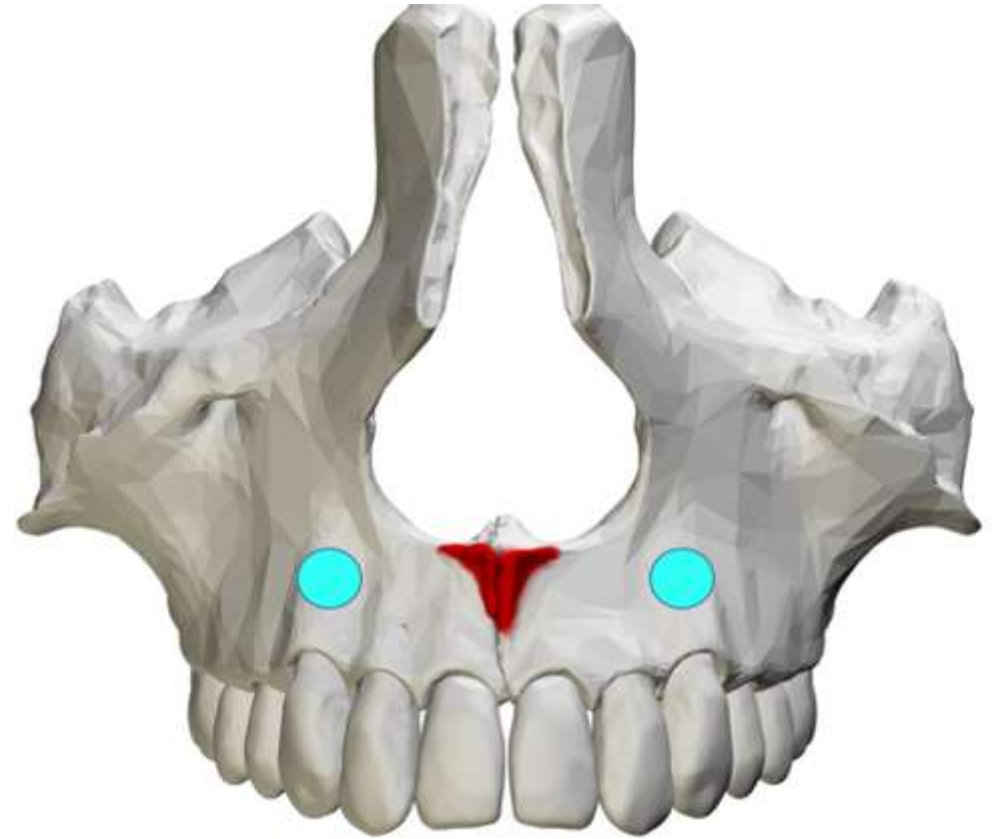
Ossification spreads as follows

- Forward → forming the alveolar process (which will contain tooth sockets)
- Upward → forming the frontal process (toward the orbit)
- Backward → forming the zygomatic process (toward the cheekbone)
- Medially → forming the palatine process (toward the nasal septum) and palate



Growth of Maxilla

Alveolar process development
Sutural growth
Bone remodelling
Enlargement of maxillary sinus
Subperiosteal bone formation



Growth of Maxilla

1. Sutural Growth: (median palatine, frontomaxillary, zygomaticomaxillary, nasomaxillary, etc.). New bone is deposited at these sutures. Pushes the maxilla **downward and forward**: increase in facial height

2. Alveolar Process Development: Growth of the alveolar process increases the **vertical height of the maxilla**.

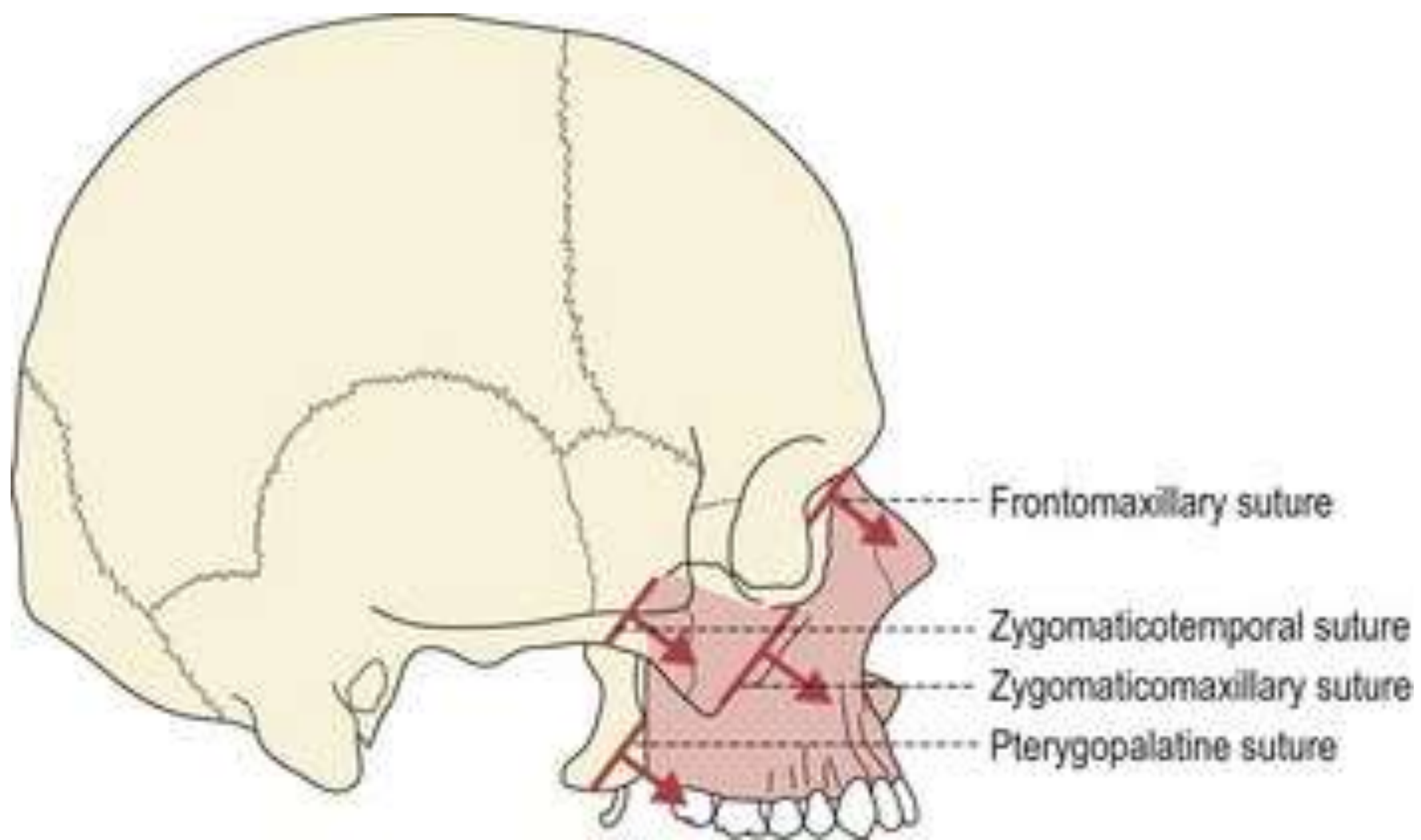
3. Bone Resorption and Formation (Remodeling): Bone is deposited on the posterior surface of the maxilla. Bone is resorbed on some anterior surfaces.

4. Enlargement of the Maxillary Sinus

The maxillary sinus enlarges during growth by resorption of its walls: growth of the midface.

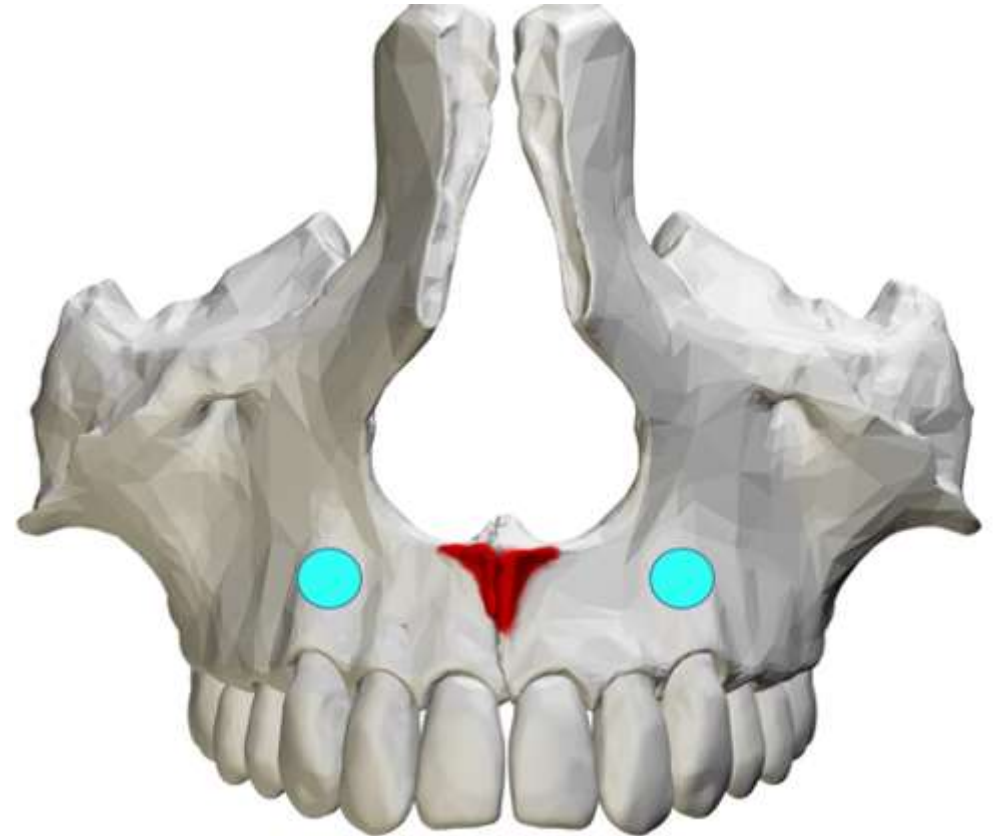
5. Subperiosteal Bone Formation

New bone is deposited beneath the periosteum. This increases the width and thickness of the maxilla.



Growth of Maxilla

- The nasal septum is thought to pull the maxilla forward through a small ligament called the septo-premaxillary ligament
- The height of the maxilla increases mainly because of the growth of the alveolar process
- Width of the maxilla increases at the median palatine suture, with more widening at the back than at the front.



THANK YOU